

Waste Classification Recycling Helper

ITAI 1378 – Computer Vision and AI

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Project Tier: Tier 1 – Core

Goal: Use computer vision to classify waste into common recycling categories.

The Problem

- People often have difficulty deciding how to properly sort household waste.
- Incorrect sorting can cause recyclable materials to be placed in the trash and can make recycling less effective.
- Who cares? Households • Schools • Businesses • Recycling programs
- Problem: People need a simple way to identify what type of waste they have.

Your Solution

- A computer vision system that identifies the category of a waste item from an image.

INPUT → MODEL → OUTPUT

- Waste image → Image classification model → Waste category
- Example: Image of plastic bottle → Plastic

Technical Approach

- CV Technique: Image Classification
- Model Architecture: CNN
- Model: MobileNet
- Framework: TensorFlow / Keras
- Approach: Transfer learning using a pretrained model
- Why: MobileNet is lightweight and appropriate for a Tier 1 project using free computing resources.

Data Plan

- Dataset: TrashNet
- Source: Public dataset
- Labels: Plastic • Paper • Metal • Glass • Cardboard • Trash
- The images will be divided into training and testing data so the model can be evaluated on images it has not seen before.
- Note: Exact dataset size and public source link will be verified before final submission.

Success Metrics

PRIMARY METRIC: Classification Accuracy

Target: At least 85% accuracy

SECONDARY METRIC: Inference Speed

Target: Under 1 second per image

Additional evaluation: Confusion matrix, misclassified images, and performance across waste categories.

Milestone Plan

Blueprint — Week 5: Complete proposal and GitHub setup

First Working Demo — Week 6: Run pretrained model on sample images

Make It Yours — Weeks 7–8: Train/test using waste dataset

Improve and Measure — Week 9: Evaluate accuracy and improve model

Build — Week 10: Complete demo, README, and final presentation

Risks and Resources

RISK 1: Model accuracy is lower than expected

PLAN B: Improve preprocessing/augmentation or use EfficientNet

RISK 2: Training or Colab problems

PLAN B: Use a pretrained model and focus on application/evaluation

Resources: Google Colab • TensorFlow/Keras • TrashNet • Kaggle Notebook

Estimated Cost: \$0